BUS 445 | STORYTELLING WITH DATA PROJECT

Dataset Description
In-Vehicle Coupon Recommendation

SFU Beedie School of Business Summer 2026 · Section D100

1. Business Context

This dataset captures a large-scale survey experiment designed to understand when drivers will accept a contextual coupon delivered through an in-vehicle recommendation system. Each respondent was presented with a series of hypothetical driving scenarios that varied along contextual dimensions such as weather, time of day, destination, who was in the car, and the type and distance of the offer. For every scenario, the respondent indicated whether they would accept the coupon if it were offered.

The unit of analysis is a single scenario presented to a single respondent, so each row is one hypothetical offer in one context. This structure lets you study how driving context and personal preferences interact to drive acceptance, which is exactly the logic a real recommendation engine would need to encode. Because the response is what people said they would do rather than what they actually did, the data reflects stated preference, not revealed preference. That distinction is central to how the findings should be interpreted and validated.

> **Stated preference, not revealed preference**
> This data records intentions expressed in a survey, not behavior observed in a real vehicle. A strong analysis names this limitation, reasons about which findings are most likely to survive contact with the real world, and proposes a validation path such as a live A/B test. The strongest recommendations distinguish a defensible launch rule from a hypothesis that still needs an experiment.

2. Data Scope & Coverage

- 12,684 scenario-level records, each representing one hypothetical coupon offer presented to one respondent.
- A single CSV file; no joins are required.
- A binary target, Y, indicating whether the coupon was accepted. Overall acceptance in the data is approximately 57 percent, so the classes are relatively balanced.
- Twenty-five predictor fields spanning driving context, coupon details, driver demographics, consumption habits, and the geospatial relationship between the driver and the redemption location.

- Predictors are overwhelmingly categorical, which shapes how the data must be encoded and modeled.

3. Understanding Key Concepts in the Data

Scenario-Level vs. Respondent-Level Variables

The variables fall into two conceptual groups. Some describe the scenario and vary from row to row for the same person: the driving context (destination, passenger, weather, temperature, time), the coupon details (type, expiration), and the geospatial factors (distance and direction to the redemption location). Others describe the respondent and are constant across all of that person's scenarios: the driver profile (gender, age, marital status, children, education, occupation, income) and the self-reported consumption habits. Recognizing which is which matters for interpretation, since the scenario variables are the levers the product can actually control, while the respondent variables describe who is being targeted.

The Consumption-Habit Variables

Five variables record how often the respondent visits bars, coffee houses, take-away outlets, and two tiers of restaurants. They are stored as ordered frequency bands using the labels never, less1 (less than once a month), 1~3, 4~8, and gt8 (more than eight times a month). Although stored as text, they carry a natural order, so you may choose to treat them as ordered categories or map them to numeric levels. These habit variables are often among the strongest predictors, since a driver's baseline behavior toward a category tends to shape their receptiveness to a coupon in that category.

Geospatial Factors

Three usable indicators capture the redemption location relative to the driver: whether it is more than 15 minutes away, more than 25 minutes away, and whether it lies in the same or the opposite direction of travel. These nested distance flags let you study how willingness to accept decays as an offer becomes less convenient. Note that a fourth field, toCoupon_GEQ5min, is present but takes the same value for every row, so it carries no information and should be dropped.

4. Data Quality Notes

A brief exploration of the file surfaces a few issues worth handling early, before any modeling. Attending to these is part of the analytical work.

- **The car field is almost entirely empty.** The car variable is missing for roughly 99 percent of records and should be dropped rather than imputed. It is included here only for transparency about what the raw file contains.
- **A constant column carries no signal.** toCoupon_GEQ5min takes the same value in every row. It has no variance and should be removed.
- **The habit variables have minor missingness.** The five consumption-habit fields each have a small number of missing values (on the order of one to two percent). Decide deliberately whether to drop those rows or treat missing as its own category, and justify the choice.
- **Watch the raw category labels.** Several fields use compact codes, for example expiration is coded 1d and 2h, and the habit bands use less1 and gt8. Clean and relabel these thoughtfully so your outputs are readable.
- **A spelling note in the header.** The passenger column is spelled passanger in the raw file. Refer to it by its exact header name when coding.

5. Data Dictionary

The complete variable-level reference for the file follows. Category values are shown as they appear in the raw data.

Variable	Type	Description
Target Variable		
Y	*binary*	1 if the coupon was accepted in the scenario; 0 otherwise. The prediction target.
Driving Context (scenario-level)		
destination	*categorical*	The driver’s destination: No Urgent Place, Home, or Work.
passanger	*categorical*	Who is in the car with the driver: Alone, Friend(s), Kid(s), or Partner. (Spelled “passanger” in the file.)
weather	*categorical*	Weather in the scenario: Sunny, Rainy, or Snowy.
temperature	*numeric*	Outside temperature in Fahrenheit, taking one of three values: 30, 55, or 80.
time	*categorical*	Time of day: 7AM, 10AM, 2PM, 6PM, or 10PM.
Coupon Details (scenario-level)		

Variable	Type	Description
coupon	*categorical*	Type of coupon offered: Coffee House, Restaurant(<20), Restaurant(20-50), Bar, or Carry out & Take away.
expiration	*categorical*	Validity window of the coupon: 1d (one day) or 2h (two hours).
Driver Profile (respondent-level)		
gender	*categorical*	The driver's gender.
age	*categorical*	Age band: below21, 21, 26, 31, 36, 41, 46, or 50plus.
maritalStatus	*categorical*	Single, Married partner, Unmarried partner, Divorced, or Widowed.
has_children	*binary*	1 if the respondent has children, 0 otherwise.
education	*categorical*	Highest education attained, from Some High School through Graduate degree.
occupation	*categorical*	Occupation grouping (25 categories, for example Student, Sales & Related, Management, Unemployed).
income	*categorical*	Annual income band, in nine ranges from Less than $12500 to $100000 or More.
car	*categorical*	Type of car driven. Missing for roughly 99% of records; recommended to drop.
Consumption Habits (respondent-level)		
Bar	*categorical (ordered)*	Monthly frequency of visiting bars: never, less1, 1~3, 4~8, gt8.
CoffeeHouse	*categorical (ordered)*	Monthly frequency of visiting coffee houses (same bands).
CarryAway	*categorical (ordered)*	Monthly frequency of getting take-away food (same bands).
RestaurantLessThan20	*categorical (ordered)*	Monthly frequency of visiting restaurants averaging under $20 per person (same bands).
Restaurant20To50	*categorical (ordered)*	Monthly frequency of visiting restaurants averaging $20 to $50 per person (same bands).
Geospatial Factors (scenario-level)		
toCoupon_GEQ5min	*binary*	Constant in this file (always 1). No variance; recommended to drop.
toCoupon_GEQ15min	*binary*	1 if the redemption location is more than 15 minutes away.
toCoupon_GEQ25min	*binary*	1 if the redemption location is more than 25 minutes away.
direction_same	*binary*	1 if the redemption location is in the same direction as the driver's travel.

Variable	Type	Description
direction_opp	*binary*	1 if the redemption location is in the opposite direction of travel.

6. Analytical Dimensions & Opportunities

This dataset supports a range of customer analytics approaches. The following dimensions highlight where analytical value can be extracted.

Contextual Acceptance Modeling

- Acceptance varies sharply by coupon type and context. In the raw data, take-away and inexpensive-restaurant offers are accepted far more often than bar offers, which points to strong category and context effects worth modeling.
- The scenario variables (weather, time, destination, passenger, distance, direction) are the levers a live engine controls, making them the highest-value predictors for a deployable rulebook.

Interaction Effects & Personalization

- The richest signal often lives in interactions, for example how a coffee coupon performs at 7AM heading to work versus 10PM heading home, or how a bar offer performs with a partner in the car versus with children.
- Consumption habits interact with context: a frequent coffee-house visitor may respond to a coffee coupon in conditions where an infrequent one would not.

Interpretable Rules vs. Predictive Power

- The case supports both an interpretable rulebook (transparent, defensible conditions under which to serve an offer) and a maximally predictive model. Choosing between, or combining, these is a central strategic decision.

Validation & Causal Caution

- Because the data is stated preference, the most valuable recommendations specify how a finding would be validated in deployment, including the design of a real-world A/B test to confirm the effect before a full rollout.

7. File Included in This Dataset

File Name	Description	Approx. Rows
in-vehicle-coupon-recommendation.csv	Scenario-level survey responses with context, coupon, driver, habit, and geospatial variables plus the acceptance outcome.	~12,684

This document is a reference guide for your team's project work. If you have questions about the data structure or variables, please raise them during class or office hours.